

Sakshi Mishra

Dept. of Civil Engineering
Rajkiya Engineering College, Kannauj

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Major Project - II (BCE851) (Project Briefing)

Research Project: Resonance-Avoidance Speed Control for Bridges: A Self-Optimizing Traffic-Structure Interaction Framework.

Resonance-Avoidance Speed Control for Bridges

(A Self-Optimizing Traffic-Structure Interaction Framework)

Bridge resonance induced by moving vehicular loads can lead to excessive vibration, accelerated fatigue damage, and reduced service life. Conventional mitigation techniques rely on structural modification such as increased stiffness or damping, which are costly and inflexible.

This project proposes a ***Resonance-Avoidance Speed Control (RASC)*** framework in which traffic speed is treated as a controllable system variable. By identifying resonance-prone speed ranges based on bridge modal properties and dynamically regulating allowable vehicle speeds, the bridge-traffic system becomes self-optimizing. **The proposed framework integrates structural dynamics, moving load theory, and control logic into a unified system architecture.** Validation is carried out through numerical simulation, demonstrating significant reduction in dynamic amplification and fatigue indicators without physical alteration of the structure.

I. KEY GOALS

1. **Resonance Prevention** – Avoid excitation frequencies that coincide with bridge natural frequencies.
2. **Non-Intrusive Control** – Eliminate the need for structural retrofitting or mechanical modification.
3. **System-Level Optimization** – Treat traffic and structure as a coupled dynamic system.

4. **Scalability** – Applicable to multiple bridge types and spans.
5. **Practical Implement ability** – Compatible with existing traffic control infrastructure.

II. SYSTEM OVERVIEW

The proposed Resonance-Avoidance Speed Control (RASC) framework operates as an integrated traffic-structure interaction system designed to minimize bridge resonance through intelligent speed regulation. Instead of relying on structural strengthening or additional damping mechanisms, the framework continuously evaluates the relationship between vehicle-induced excitation frequencies and the bridge's natural frequencies to identify potentially unsafe operating conditions.

The system begins with structural dynamic analysis of the bridge, where modal characteristics such as natural frequencies and dominant mode shapes are determined using finite element modeling and eigenvalue analysis. Simultaneously, moving vehicular loads are modeled to evaluate how different vehicle speeds influence dynamic excitation of the structure.

Using this interaction data, the framework generates speed-response relationships and identifies resonance-prone speed intervals known as resonance

zones. These zones represent operating conditions where dynamic amplification, vibration levels, or fatigue stresses exceed predefined safety thresholds.

A control module is then employed to regulate allowable vehicle speeds by restricting resonance-prone speed bands and promoting safe operating ranges. The control logic may operate using predefined rule-based constraints or optimization algorithms that balance structural safety with traffic efficiency.

III. STRUCTURAL DYNAMICS MODEL

Bridge Representation

The bridge is modeled as a linear elastic beam or girder system governed by:

$$M\ddot{x}(t) + C\dot{x}(t) + Kx(t) = F(t)$$

Where:

- M, C, K are the mass, damping, and stiffness matrices respectively
- F(t) is the moving vehicle load

Modal Analysis

Natural frequencies (f_n) and mode shapes (ϕ_n) are obtained via eigenvalue analysis. Only the first few vertical bending modes are considered, as they dominate resonance behavior.

IV. TRAFFIC EXCITATION MODEL

Moving Load Representation

Vehicles are modeled as moving point loads or distributed axle loads traveling at constant speed v .

Excitation Frequency

The fundamental excitation frequency is approximated by:

$$f_{exc} = v / L$$

Where:

- v = vehicle speed
- L = bridge span

Resonance occurs when $f_{exc} \approx f_n$.

V. RESONANCE ZONE IDENTIFICATION

Definition

A **Resonance Zone** is defined as a speed interval $[v_1, v_2]$ for which dynamic response exceeds a predefined threshold.

Identification Procedure

- Sweep vehicle speed across operating range
- Compute dynamic amplification factor (DAF)
- Identify peak response regions
- Mark corresponding speed intervals as resonance zones

VI. SPEED CONTROL ARCHITECTURE

Control Philosophy

Instead of modifying the structure, the system **modifies traffic behavior**.

Control Law

If $v \in \text{Resonance Zone} \rightarrow$ enforce speed adjustment

Allowed Speed Bands

The total speed range is partitioned into:

- Safe operating bands
- Restricted resonance bands

Optimization Objective

Minimize:

- Peak acceleration
- Stress range

- Fatigue damage index

Subject to:

- Traffic flow constraints
- Speed limit regulations

VII. IMPLEMENTATION METHODOLOGY

Structural Modeling

- Develop FEM model using ETABS / SAP2000 / OpenSees
- Perform modal and time-history analysis

Moving Load Simulation

- Apply moving load at varying speeds
- Record deflection, acceleration, and stress

Resonance Mapping

- Generate speed–response curves
- Identify resonance zones

Control Strategy Design

- Define adaptive speed rules
- Implement rule-based or optimization-based logic in MATLAB

Performance Evaluation

Compare uncontrolled vs-controlled scenarios based on:

- Dynamic amplification
- Fatigue damage
- Traffic efficiency

VIII. PERFORMANCE METRICS

- Maximum deflection
- Peak acceleration
- Dynamic Amplification Factor (DAF)
- Fatigue damage index (Miner's Rule)
- Traffic throughput impact

IX. EXPECTED OUTCOMES

- Clear identification of resonance-prone speed ranges
- Reduction in vibration amplitude (20–50%)

- Improved fatigue life without structural intervention
- Demonstration of traffic as a controllable excitation source

X. NOVEL CONTRIBUTIONS

- Introduces speed regulation as a structural control mechanism
- Establishes a traffic–structure co-optimization framework
- Provides a low-cost alternative to traditional retrofitting

XI. TOOLS AND SOFTWARE

- ETABS / SAP2000 / OpenSees – Structural analysis
- MATLAB – Control logic and optimization
- Python (optional) – Parametric analysis

XII. LIMITATIONS AND EXTENSIONS

Limitations

- Linear elastic assumption
- Simplified vehicle model

Future Extensions

- Real-time sensor integration
- Multi-lane traffic coupling
- Fatigue-based life-cycle optimization

XIII. CONCLUSION

The Resonance-Avoidance Speed Control framework demonstrates that bridge resonance can be mitigated effectively through intelligent traffic management. By reinterpreting traffic speed as a control variable, the proposed SOTSS introduces a new paradigm in smart and resilient infrastructure design.